Section 3.2 IM-OLS Cointegration Estimation Matrix: Specification B Heterogeneous Capital (1929–2024)

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Data Provenance and Estimation Summary

This report presents the empirical cointegration results for Section 3.2 of Chapter 2 (Specification B). All models are estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang et al. 2024) over the full NIPA sample window (1931–2024, $T = 94$ observations after listwise deletion for lag-1 conditioning variables). Capital is disaggregated into non-residential structures (k^{NRC}) and the mechanization gap ($\kappa_t = k^{\text{ME}} - k^{\text{NRC}}$). To resolve multi-collinearity, all interaction terms are orthogonalized via Frisch-Waugh-Lovell (FWL) against base capital regressors and any detected Bai-Perron step dummies.

Theoretical Rationale: Binding Net Value Added (NVA) & Distributive Domain

Under the synthesis of Okishio's market anarchy and Vidal's internally divided firm, class struggle over distribution is defined strictly over newly created value (Net Value Added $= v + s$), excluding pre-existing constant capital transfers (CFC, or physical depreciation):

$$\begin{aligned} \text{Gross Output } (W) &= c + v + s \\ \text{Gross Value Added (GVA)} &= CFC + v + s \\ \text{Net Value Added (NVA)} &= v + s \end{aligned}$$

1. **Exclusion of Constant Capital Transfer (CFC):** Physical depreciation (CFC) represents pre-existing value transferred to output by concrete labor. Because CFC is not new value created on the shop floor, it lies outside zero-sum class struggle over surplus distribution.

2. **Elimination of Mechanization Distortion:** During intensive mechanization (substituting machinery for living labor), the ratio CFC/GVA rises mechanically. Assessing wage share over GVA ($\omega_{\text{GVA}} = \frac{\text{COMP}}{CFC+v+s}$) causes the wage share to appear to fall even if the rate of surplus value extraction (s/v) remains constant. Evaluating wage share over NVA ($\omega_{\text{NFC,NVA}} = \frac{\text{COMP}}{\text{NFC_NVA}}$) isolates the true balance of class power without pollution from physical depreciation shifts.

3. **Supervisory Adjustment (Shaikh-Tonak Tier 1 & Tier 2):** Standard NIPA compensation includes managerial and executive salaries, which function as a distribution of surplus value rather than a variable labor cost. The Shaikh-Tonak Tier 1 ($\omega_{\text{ST,Tier1}} = \frac{\text{COMP}}{\text{NFC_NVA}}$) and Tier 2 ($\omega_{\text{ST,Tier2}} = \frac{v_{\text{prod}}}{\text{NFC_NVA}}$) measures strip supervisory overhead to isolate true variable capital (v), capturing the precise variable labor cost that enters corporate cost minimization decisions.

4. **Clean Corporate Sector Adjustments:** To ensure maximum robustness, we present an additional set of wage proxies ("Clean Corporate Tier 1 & Tier 2") that precisely bound the non-financial corporate sector to eliminate spurious non-profit/proprietor income flows that contaminate standard NFC aggregate data. These series are strictly bounded to the empirical support window (Tier 2 $T = 77$) without imposing constant-share back-extrapolation.

Key Empirical Findings and Cointegration Verdict

1. **Cointegration Triad Verdict:** All estimated models reject the null hypothesis of no cointegration under the Engle-Granger Residual ADF test ($t_{\text{ADF}} \leq -4.19, p < 0.05$). Crucially, after applying FWL detrending to the deterministic space and correcting the Shin LM accumulator, both the Shin LM test ($C_{\text{Shin}} \leq 0.057$) and the Vogelsang-Wagner Type 2 Wald test ($S_{\text{Type 2}} \leq 15.63$) fail to reject the null hypothesis of cointegration across all specifications. This provides robust mutual confirmation of stable cointegration under IM-OLS, confirming that structural break step-shifts absorb historical regime shifts without spurious trend contamination.

2. **Failure of Unadjusted GVA Wage Share (Set 1):** In standard interactive specifications (Models M2.2a/b and M2.4a/b), the unadjusted wage interaction ($\hat{\theta}_{\omega\kappa}$) is statistically insignificant ($t = 0.54$ for M2.2a; $t = -1.00$ for M2.2b), demonstrating that raw wage shares fail to capture accumulation-channel dynamics.

3. **Success of NVA-Based Effective Wage Pressure (Set 2):** When scaled by productive accumulation ($1 - \iota_{t-1}$) in Set 2, the effective wage interaction ($\hat{\theta}_{\omega\kappa}$) becomes statistically significant with the theoretically predicted negative sign ($\hat{\theta}_{\omega\kappa} = -15.99, t = -3.24, p < 0.01$ in M2.6a; $\hat{\theta}_{\omega\kappa} = -12.29, t = -2.91, p < 0.05$ in M2.6b; $\hat{\theta}_{\omega\kappa} = -12.77, t = -3.36, p < 0.01$ in M2.8a). High effective wage pressure forces defensive mechanization, increasing the organic composition of capital and depressing marginal capacity yield.

1 Specification Set 1: Standard Wage Interaction (GVA vs. NVA Comparison)

1.1 Panel A: Standard Wage Interaction over Gross Value Added (GVA Basis)

Table 1: Specification Set 1: Standard Wage Interaction (GVA Basis)

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)										
Model	M2.1a	M2.2b	M2.5a	M2.6b	M2.9a	M2.10b	M2.13a	M2.14b	M2.17a	M2.18b
$\hat{\alpha}$ (Intercept)	12.90***	12.82***	12.90***	12.82***	13.85	13.52	12.89***	12.81***	13.85	13.52
	(0.07)	(0.06)	(0.07)	(0.06)	(0.08)	(0.06)	(0.07)	(0.07)	(0.08)	(0.06)
$\hat{\theta}_{NRC}$ (Structures k^{NRC})	0.60***	0.51***	0.61***	0.48***	0.51	0.50	0.59***	0.52***	0.51	0.49
	(0.03)	(0.08)	(0.03)	(0.07)	(0.02)	(0.07)	(0.03)	(0.09)	(0.02)	(0.07)
$\hat{\theta}_{\kappa}$ (Mech. Gap κ)	0.38**	0.67**	0.39**	0.53*	0.38	1.09	0.36**	0.60	0.34	1.11
	(0.14)	(0.26)	(0.14)	(0.23)	(0.12)	(0.20)	(0.14)	(0.33)	(0.11)	(0.20)
$\hat{\theta}_{\omega\kappa}$ (FWL Wage Interact)	-2.75	-4.89	4.01	-4.20	-16.40	-10.51	-9.89*	-5.93	-18.27	-9.04
	(4.07)	(3.10)	(7.41)	(4.18)	(6.63)	(5.86)	(4.85)	(3.96)	(5.56)	(5.55)
$\hat{\theta}_{\kappa p}$ (FWL Price Interact)	-0.68	1.38	-1.28*	0.64	-1.43	-0.35	0.01	1.63	-0.78	-0.35
	(0.58)	(0.82)	(0.61)	(0.65)	(0.56)	(0.47)	(0.57)	(0.99)	(0.54)	(0.47)
Step Shifts ($D_{T_b}^{\mathrm{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Collinearity Diagnostics (VIF)										
VIF (k^{NRC})	1.69	1.69	1.69	1.69	1.02	1.02	1.69	1.69	1.02	1.02
VIF (κ)	1.69	1.69	1.69	1.69	1.02	1.02	1.69	1.69	1.02	1.02
VIF (Wage Interaction)	1.32	1.06	1.15	1.01	1.03	1.00	1.34	1.15	1.09	1.00
VIF (Price Interaction)	1.32	1.06	1.15	1.01	1.03	1.00	1.34	1.15	1.09	1.00
Cointegration Test Triad										
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.16*	-4.46**	-4.26*	-4.69**	-2.56	-3.22	-4.26*	-4.50**	-2.58	-3.17
Shin LM (C_{Shin}) [H_0: Coint.]	0.05	0.03	0.05	0.03	0.03	0.02	0.05	0.03	0.02	0.02
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	16.11	9.55	16.25	11.41	9.98	2.78	16.37	10.87	8.42	3.26
Bandwidth Ratio (b)	0.2088	0.1758	0.2114	0.1744	0.2112	0.2029	0.2128	0.1906	0.2098	0.2028
Sample Size (T)	94	94	94	94	76	76	94	94	76	76

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. Stars for coefficient rows (structures, mechanization gap, and interactions) and the Residual VW Type 2 cointegration test are derived from a 100,000-pass custom Monte Carlo simulation suite mapping the exact DGP variant (with/without step level-shifts). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies.

1.2 Panel B: Standard Wage Interaction over Net Value Added (NVA Basis)

Table 2: Specification Set 1: Standard Wage Interaction (NVA Basis)

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)										
Model	M2.3a	M2.4b	M2.7a	M2.8b	M2.11a	M2.12b	M2.15a	M2.16b	M2.19a	M2.20b
$\hat{\alpha}$ (Intercept)	12.90***	12.82***	12.90***	12.82***	13.87	13.57	12.90***	12.82***	13.88	13.57
	(0.07)	(0.06)	(0.07)	(0.06)	(0.08)	(0.06)	(0.07)	(0.07)	(0.08)	(0.06)
$\hat{\theta}_{NRC}$ (Structures k^{NRC})	0.60***	0.51***	0.60***	0.51***	0.51	0.52	0.59***	0.50***	0.51	0.50
	(0.03)	(0.08)	(0.03)	(0.08)	(0.02)	(0.07)	(0.03)	(0.09)	(0.02)	(0.07)
$\hat{\theta}_{\kappa}$ (Mech. Gap κ)	0.38**	0.67**	0.38**	0.67**	0.35	0.99	0.36**	0.61	0.34	1.02
	(0.14)	(0.26)	(0.14)	(0.26)	(0.11)	(0.18)	(0.14)	(0.34)	(0.11)	(0.18)
$\hat{\theta}_{\omega\kappa}$ (FWL Wage Interact)	-2.75	-4.89	-2.75	-4.89	-13.06	-13.10	-7.06*	-2.73	-12.13	-11.13
	(4.07)	(3.10)	(4.07)	(3.10)	(4.24)	(4.38)	(3.32)	(2.62)	(3.50)	(3.99)
$\hat{\theta}_{\kappa p}$ (FWL Price Interact)	-0.68	1.38	-0.68	1.38	-0.99	-0.37	0.33	1.50	-0.49	-0.30
	(0.58)	(0.82)	(0.58)	(0.82)	(0.52)	(0.42)	(0.65)	(0.98)	(0.53)	(0.43)
Step Shifts ($D_{T_b}^{\mathrm{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Collinearity Diagnostics (VIF)										
VIF (k^{NRC})	1.69	1.69	1.69	1.69	1.02	1.02	1.69	1.69	1.02	1.02
VIF (κ)	1.69	1.69	1.69	1.69	1.02	1.02	1.69	1.69	1.02	1.02
VIF (Wage Interaction)	1.32	1.06	1.32	1.06	1.06	1.00	1.56	1.18	1.13	1.00
VIF (Price Interaction)	1.32	1.06	1.32	1.06	1.06	1.00	1.56	1.18	1.13	1.00
Cointegration Test Triad										
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.16*	-4.46**	-4.16*	-4.46**	-2.61	-3.55	-4.20*	-4.49**	-2.50	-3.44
Shin LM (C_{Shin}) [H_0: Coint.]	0.05	0.03	0.05	0.03	0.02	0.02	0.05	0.03	0.02	0.02
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	16.11	9.55	16.11	9.55	8.41	6.15	16.35	11.90	7.50	4.26
Bandwidth Ratio (b)	0.2088	0.1758	0.2088	0.1758	0.2072	0.2053	0.2128	0.1918	0.2110	0.2048
Sample Size (T)	94	94	94	94	76	76	94	94	76	76

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. Stars for coefficient rows (structures, mechanization gap, and interactions) and the Residual VW Type 2 cointegration test are derived from a 100,000-pass custom Monte Carlo simulation suite mapping the exact DGP variant (with/without step level-shifts). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies.

2 Specification Set 2: Effective Wage Interaction (GVA vs. NVA Comparison)

2.1 Panel A: Effective Wage Interaction over Gross Value Added (GVA Basis)

Table 3: Specification Set 2: Effective Wage Interaction (GVA Basis)

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)										
Model	M2.21a	M2.22b	M2.25a	M2.26b	M2.29a	M2.30b	M2.33a	M2.34b	M2.37a	M2.38b
$\hat{\alpha}$ (Intercept)	12.82^{***}	12.77^{***}	12.79^{***}	12.76^{***}	13.68	13.66	12.80^{***}	12.77^{***}	13.67	13.61
	(0.07)	(0.06)	(0.07)	(0.06)	(0.06)	(0.05)	(0.07)	(0.06)	(0.05)	(0.06)
$\hat{\theta}_{NRC}$ (Structures k^{NRC})	0.61^{***}	0.58^{***}	0.61^{***}	0.62^{***}	0.55	0.51	0.61^{***}	0.60^{***}	0.55	0.70
	(0.02)	(0.08)	(0.02)	(0.08)	(0.01)	(0.01)	(0.02)	(0.09)	(0.01)	(0.03)
$\hat{\theta}_{\kappa}$ (Mech. Gap κ)	0.32^{**}	0.39	0.31^{*}	0.14	0.36	0.02	0.31^{*}	0.36	0.34	0.72
	(0.13)	(0.29)	(0.14)	(0.38)	(0.08)	(0.11)	(0.14)	(0.29)	(0.08)	(0.10)
$\hat{\theta}_{\omega\kappa}$ (FWL Wage Interact)	- 12.77^{***}	-10.13^{**}	- 15.99^{***}	-12.29^{**}	-20.77	-19.87	- 14.54^{***}	-11.15^{**}	-18.85	-15.55
	(3.81)	(3.43)	(4.94)	(4.22)	(4.11)	(3.99)	(4.60)	(3.63)	(3.32)	(3.54)
$\hat{\theta}_{\kappa p}$ (FWL Price Interact)	0.18	0.92	-0.39	0.58	-1.37	-0.81	0.08	0.88	-0.96	0.10
	(0.51)	(0.69)	(0.45)	(0.60)	(0.42)	(0.39)	(0.52)	(0.66)	(0.40)	(0.41)
Step Shifts ($D_{T_b}^{\mathrm{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Collinearity Diagnostics (VIF)										
VIF (k^{NRC})	1.69	1.69	1.69	1.69	1.02	1.02	1.69	1.69	1.02	1.02
VIF (κ)	1.69	1.69	1.69	1.69	1.02	1.02	1.69	1.69	1.02	1.02
VIF (Wage Interaction)	1.18	1.03	1.03	1.00	1.05	1.02	1.15	1.04	1.12	1.06
VIF (Price Interaction)	1.18	1.03	1.03	1.00	1.05	1.02	1.15	1.04	1.12	1.06
Cointegration Test Triad										
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.33*	-4.58^{**}	-4.40*	-4.77^{**}	-4.12	-4.02	-4.41*	-4.69^{**}	-4.13*	-3.97
Shin LM (C_{Shin}) [H_0: Coint.]	0.03	0.02	0.03	0.02	0.03	0.02	0.03	0.02	0.03	0.02
VW Type 2 ($S_{\mathrm{Type\ 2}}$) [H_0: Coint.]	14.97	5.93	14.58	3.71	13.20	11.19	14.75	6.00	12.33	9.93
Bandwidth Ratio (b)	0.1854	0.1731	0.1921	0.1735	0.2063	0.2065	0.1948	0.1790	0.2070	0.2069
Sample Size (T)	94	94	94	94	76	76	94	94	76	76

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. Stars for coefficient rows (structures, mechanization gap, and interactions) and the Residual VW Type 2 cointegration test are derived from a 100,000-pass custom Monte Carlo simulation suite mapping the exact DGP variant (with/without step level-shifts). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies.

2.2 Panel B: Effective Wage Interaction over Net Value Added (NVA Basis)

Table 4: Specification Set 2: Effective Wage Interaction (NVA Basis)

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)										
Model	M2.23a	M2.24b	M2.27a	M2.28b	M2.31a	M2.32b	M2.35a	M2.36b	M2.39a	M2.40b
$\hat{\alpha}$ (Intercept)	12.82***	12.77***	12.82***	12.77***	13.72	13.69	12.83***	12.78***	13.73	13.73
	(0.07)	(0.06)	(0.07)	(0.06)	(0.04)	(0.04)	(0.07)	(0.07)	(0.04)	(0.04)
$\hat{\theta}_{NRC}$ (Structures k^{NRC})	0.61***	0.58***	0.61***	0.58***	0.54	0.51	0.60***	0.60***	0.54	0.54
	(0.02)	(0.08)	(0.02)	(0.08)	(0.01)	(0.01)	(0.02)	(0.08)	(0.01)	(0.01)
$\hat{\theta}_{\kappa}$ (Mech. Gap κ)	0.32**	0.39	0.32**	0.39	0.33	0.06	0.32**	0.39	0.31	0.31
	(0.13)	(0.29)	(0.13)	(0.29)	(0.06)	(0.09)	(0.13)	(0.29)	(0.06)	(0.06)
$\hat{\theta}_{\omega\kappa}$ (FWL Wage Interact)	-12.77***	-10.13**	-12.77***	-10.13**	-17.05	-16.76	-11.08***	-7.93**	-14.97	-14.97
	(3.81)	(3.43)	(3.81)	(3.43)	(2.34)	(2.55)	(3.48)	(2.96)	(2.01)	(2.01)
$\hat{\theta}_{\kappa p}$ (FWL Price Interact)	0.18	0.92	0.18	0.92	-0.79	-0.71	0.69	1.41	-0.31	-0.31
	(0.51)	(0.69)	(0.51)	(0.69)	(0.32)	(0.31)	(0.63)	(0.75)	(0.33)	(0.33)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Collinearity Diagnostics (VIF)										
VIF (k^{NRC})	1.69	1.69	1.69	1.69	1.02	1.02	1.69	1.69	1.02	1.02
VIF (κ)	1.69	1.69	1.69	1.69	1.02	1.02	1.69	1.69	1.02	1.02
VIF (Wage Interaction)	1.18	1.03	1.18	1.03	1.10	1.03	1.42	1.10	1.20	1.20
VIF (Price Interaction)	1.18	1.03	1.18	1.03	1.10	1.03	1.42	1.10	1.20	1.20
Cointegration Test Triad										
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.33*	-4.58**	-4.33*	-4.58**	-4.71**	-4.57**	-4.34*	-4.60**	-4.24*	-4.24*
Shin LM (C_{Shin}) [H_0: Coint.]	0.03	0.02	0.03	0.02	0.02	0.02	0.03	0.03	0.02	0.02
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	14.97	5.93	14.97	5.93	10.28	8.70	15.37	6.37	8.96	8.96
Bandwidth Ratio (b)	0.1854	0.1731	0.1854	0.1731	0.2083	0.2082	0.1929	0.1809	0.2080	0.2080
Sample Size (T)	94	94	94	94	76	76	94	94	76	76

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. Stars for coefficient rows (structures, mechanization gap, and interactions) and the Residual VW Type 2 cointegration test are derived from a 100,000-pass custom Monte Carlo simulation suite mapping the exact DGP variant (with/without step level-shifts). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies.